

Which physical issue is related to unpredictable behavior due to timing and physical issues in RTL design?

- ☒ Power consumption
- ☐ X-Propagation
- ☐ Performance optimization
- ☐ Area reduction

X-Propagation is related to unpredictable behavior due to timing and physical issues in RTL design.

Incorrect



Which factors determine Static Power consumption in CMOS ICs?

(Select all that apply)

- ☒ Switching threshold
- ☐ Supply voltage
- ☐ Transistor dimensions
- ☐ Clock frequency
- ☐ Signal activity

Static Power consumption is a function of the technology you are using, specifically the supply voltage, the switching threshold, and the transistor dimensions.

Incorrect



You have correctly answered 26 of 100 questions

Question 3 of 100

Metastability can be completely eliminated in RTL design.

- ☒ True
☐ False

Metastability cannot be completely eliminated; its effect can only be minimized.

Incorrect



You have correctly answered 26 of 100 questions

Question 4 of 100

Which of the following are valid format specifiers in Verilog for displaying values?

(Select all that apply)

- ☒ %d
☐ %b
☐ %h
☐ %o

These format specifiers are used to display values in binary, octal, decimal, and hexadecimal formats, respectively.

Incorrect



Dynamic power consumption in CMOS ICs is constant and cannot be reduced through RTL coding techniques.

- ☒ True
- ☐ False

Dynamic power consumption in CMOS ICs can be reduced through RTL coding techniques such as clock gating.

Incorrect



What is the primary purpose of clock domain crossing (CDC) verification?

- ☒ To minimize metastability
- ☐ To increase performance
- ☐ To optimize power consumption
- ☐ To reduce area

CDC verification aims to minimize metastability when signals cross from one asynchronous clock domain to another.

Correct



You have correctly answered 26 of 100 questions

Question 7 of 100

Tcl variables store only integer values.

- ☒ True
- ☐ False

Tcl variables store string values, which can represent integer, real, Boolean values, and data structures such as lists and arrays.

Incorrect



You have correctly answered 26 of 100 questions

Question 8 of 100

The ``ifdef` directive is used for conditionally compiling code based on the existence of a text macro.

- ☒ True
- ☐ False

The ``ifdef` directive conditionally compiles code based on the existence of a text macro.

Correct



You have correctly answered 26 of 100 questions

Question 9 of 100

What is the primary difference between blocking and nonblocking assignments in Verilog?

- ☒ Nonblocking assignments update immediately.
- ☐ Blocking assignments update immediately.
- ☐ Nonblocking assignments are used only for combinational logic.
- ☐ Blocking assignments schedule completion and do not update immediately.

Blocking assignments block further execution until the assignment completes, while nonblocking assignments schedule completion and do not block further execution.

Incorrect



You have correctly answered 26 of 100 questions

Question 10 of 100

Select all the unary reduction operators in Verilog: (Select all that apply)

- ☒ ~&
- ☐ ~|
- ☐ |
- ☐ &

Unary reduction operators include &, ~&, |, and ~|, which perform a bitwise operation across all bits of a single operand.

Incorrect



What will be the value of `byte` after executing the following code snippet?

- ☒ 8'b00001111
- ☐ 8'b00000000
- ☐ 8'b11111111
- ☐ 8'b11110000

```
reg [7:0] byte = 8'b00001111;  
initial begin  
    byte[3:0] <= by  
    byte[7:4] <= by  
end
```

The nonblocking assignments schedule the updates, allowing the upper half to be assigned to the lower half and vice versa, resulting in `byte` being 8'b11110000.

Incorrect

The `$monitor` system task can be disabled and re-enabled using `$monitoroff` and `$monitoron` system tasks.

- ☒ True
- ☐ False

You can disable monitoring with `$monitoroff` and re-enable it with `$monitoron`.

Correct

If clk has a positive edge and rst = 1, what will be the value of count after executing the following code snippet?

- ☒ 10
- ☐ 9
- ☐ 0
- ☐ 1

```
always @(posedge clk or posedge rst)
if (rst)
    count <= 4'd0;
else if (count == 9)
    count <= 4'd0;
else
    count <= count + 4'd1;
```

The value of count will be reset to 0 because rst is 1.

Incorrect



What is the effect of using the wildcard event control (@*) in Verilog?

- ☒ It includes only temporary variables in the event list.
- ☐ It includes all inputs to the logic in the event list.
- ☐ It excludes temporary variables from the event list.
- ☐ It excludes all inputs to the logic from the event list.

The wildcard event control (@*) ensures that all inputs to the procedure are included in the event list, avoiding incorrect RTL simulation results.

Incorrect



Which X-Optimization mode in Xcelium is known as X-Pessimism?

- ☒ Compute-As-Ternary (CAT) mode
- ☐ Optimistic mode
- ☐ Default mode
- ☐ Forward-Only-X (FOX) mode

The Forward-Only-X (FOX) mode assigns the unknown value to variables if the condition becomes unknown, and is also known as X-Pessimism mode.

Incorrect



The casex statement treats x, z, and ? characters as don't-care bit values.

- ☒ True
- ☐ False

The casex statement treats x, z, and ? characters as don't-care bit values.

Correct



What is the advantage of using Gray encoding in FSM optimization?

- ☒ Simplifies timing constraints
- ☐ Reduces power consumption
- ☐ Increases noise
- ☐ Increases register count

Gray encoding is used to reduce switching activity, which in turn reduces power consumption and noise.

Incorrect

What will be the value of a and b after executing the following code snippet?

- ☒ a = 1, b = 0.
- ☐ a = 1, b = 1.
- ☐ a = 0, b = 0.
- ☐ a = 0, b = 1.
- ☐ a = 1, the value of b will be unpredictable.

```
initial begin
    a = 0;
    b = 0;
end
always @(posedge clock)
    a = a + 1;
always @(posedge clock)
    b = a;
```

The nonblocking assignments schedule the updates, ensuring b gets the old value of a, before a is incremented.

Correct

Functions in Verilog can contain event expressions and delays.

- ☒ True
- ☐ False

Functions cannot invoke the scheduler in any way.

Incorrect

What are the two sources of power consumption in CMOS ICs?

- ☒ Inductive power and dynamic power
- ☐ Resistive power and leakage power
- ☐ Static power and dynamic power
- ☐ Capacitive power and switching power

CMOS ICs have two sources of power consumption: static power (leakage) and dynamic power (switching).

Incorrect

What is the purpose of using asynchronous reset assertion and synchronous reset de-assertion?

- ☒ To increase performance
- ☐ To minimize metastability
- ☐ To reduce power consumption
- ☐ To generate clocks

Using asynchronous reset assertion and synchronous reset de-assertion minimizes the possibility of metastability.

Incorrect



Which of the following are necessary for purely combinational logic procedures?
(Select all that apply)

- ☒ Include posedge or negedge events in the event list
- ☐ Use nonblocking assignments
- ☐ Provide default assignments to prevent latch inference
- ☐ Start the procedure with the always construct

To describe purely combinational logic, start the procedure with the always construct and provide default assignments to prevent latch inference. Use blocking assignments and avoid posedge or negedge events in the event list.

Incorrect



What does the synthesis tool infer from the HDL source?

- ☒ Testbenches
- ☐ Storage elements and combinational logic
- ☐ Only combinational logic
- ☐ Only storage elements

The synthesis tool infers storage elements (latches, registers) and combinational logic feeding storage elements.

Incorrect



Which technique is used in synchronous circuits to reduce dynamic power consumption?

- ☒ Voltage scaling
- ☐ Clock gating
- ☐ Frequency scaling
- ☐ Power shutoff

Clock gating is used in synchronous circuits to reduce dynamic power consumption by removing the clock when it is not essential to design functionality.

Incorrect



What are the benefits of using automation in verification? (Select all that apply)

- ☒ Collation and analysis of results
- ☐ Tracking of data over time
- ☐ Automatic test selection
- ☐ Manual test execution

Automation allows automatic test selection, collation and analysis of results, and tracking of data over time.

Incorrect



Select all the scenarios where blocking assignments are appropriate:
(Select all that apply)

- ☒ For all assignments in procedures representing combinational logic
- ☐ For assignments to variables representing registers in sequential procedures
- ☐ For all assignments in procedures representing sequential logic
- ☐ For assignments to temporary variables within sequential procedures

Blocking assignments are appropriate in procedures representing combinational logic and for temporary variables within sequential procedures.

Incorrect



The case statement in Verilog uses logical equality comparison.

- ☒ True
☐ False

The case statement uses case equality comparison, which considers high-impedance (z) and unknown (x) values.

Incorrect



If enable1 = 1, data_1 = 8'b10101010, data_2 = 8'b01010101 and enable2 = 0, what will be the value of data_bus after executing the following code snippet?

- ☒ 8'b1
☐ 8'bz
☐ 8'b0
☐ 8'bx
☐ 8'b10101010
☐ 8'b01010101

```
reg data_bus;  
always @*  
begin  
    if (enable1)  
        data_bus = data_1;  
    else  
        data_bus = 8'bz;  
        if (enable2)  
            data_bus = data_2;  
        else  
            data_bus = 8'bz;  
    end
```

The value of data_bus will be updated to 8'bz because the second condition (enable2 = 0) sets it to high-impedance.

Incorrect



What are the benefits of using hierarchical names in Verilog testbenches?

(Select all that apply)

- ☒ Monitoring internal signals of the DUT
- ☐ Generating clocks
- ☐ Accessing identifiers in other modules
- ☐ Complicating the testbench code

Hierarchical names allow access to identifiers in other modules and monitoring internal signals of the DUT.

Incorrect



What type of assignment should be used for variables representing registers in sequential logic?

- ☒ Continuous assignment
- ☐ Nonblocking assignment ($\leq$)
- ☐ Net declaration assignment
- ☐ Blocking assignment ($=$)

Nonblocking assignments should be used for variables representing storage elements to avoid clock/data races in simulation.

Incorrect



Which of the following statements about functions in Verilog are true?

(Select all that apply)

- ☒ They return a single value through their name
- ☐ They are invoked as a procedural statement
- ☐ They can contain timing controls
- ☐ They must have at least one input argument

Functions return a single value through their name and must have at least one input port. They cannot contain timing controls and are invoked as an expression term.

Incorrect



Which of the following is NOT a typical design constraint for synthesis?

- ☒ Color of the wires
- ☐ Output load
- ☐ Clock speed
- ☐ Input drive strength

Typical design constraints for synthesis include clock speed, input drive strength, and output load, but not the color of the wires.

Correct



What are the common causes of X values in simulation? (Select all that apply)

- ☒ Deliberate assignment in RTL code
- ☐ Unknown inputs
- ☐ Unresolvable conflict of equal-strength 0 and 1 values
- ☐ Initial uninitialized value

X values can arise from initial uninitialized values, unresolvable conflicts, unknown inputs, and deliberate assignments in RTL code.

Incorrect



Which of the following are common methods for generating stimulus in Verilog?
(Select all that apply)

- ☒ Stimulus generation using tasks
- ☐ Stimulus generation using loops
- ☐ In-line stimulus generation
- ☐ Stimulus generation from waveform patterns

Common methods for generating stimulus include in-line stimulus generation, using loops, and using tasks.

Incorrect



The repeat loop in Verilog executes its following statement a specified number of times.

- ☒ True
- ☐ False

The repeat loop executes its following statement the number of times specified by its parenthesized expression.

Correct

What is the purpose of the always construct in Verilog?

- ☒ To declare variables
- ☐ To start execution at the start of the simulation and execute repeatedly
- ☐ To perform arithmetic operations
- ☐ To start execution at the start of the simulation and terminate the process

The always construct starts execution at the start of simulation and executes repeatedly.

Incorrect

What does the 'X' value represent in simulation?

- ☒ A don't care value
- ☐ A fixed value
- ☐ An unknown value
- ☐ A known value

In simulation, the 'X' value represents an unknown value, which could be either 0 or 1.

Incorrect



Which system task would you use to open a file for writing in Verilog?

- ☒ \$fopen
- ☐ \$fmonitor
- ☐ \$fclose
- ☐ \$fwrite

The \$fopen system function opens a file and returns a 32-bit unsigned multichannel descriptor (MCD) associated with the file.

Correct



The forever loop in Verilog executes its following statement indefinitely.

- ☒ True
- ☐ False

The forever loop executes its following statement indefinitely or until simulation ends.

Correct



What is the effect of X-Optimism in RTL simulation?

- ☒ Converts unknown values to known values
- ☐ Increases power consumption
- ☐ Converts known values to unknown values
- ☐ Improves performance

X-Optimism in RTL simulation converts unknown values to known values, potentially masking design problems.

Correct



Select all the characteristics of tasks in Verilog: (Select all that apply)

- ☒ Can have zero or more input/output arguments
- ☐ Can be declared outside a module
- ☐ Can contain timing controls
- ☐ Are invoked as an expression term

Tasks can contain timing controls and can have zero or more input/output ports. They are invoked as procedural statements and must be declared within a module.

Incorrect



Which operator is used for concatenation in Verilog?

- ☒ { }
- ☐ <<
- ☐ &
- ☐ []

The concatenation operator { } joins together the bits of one or more operands to form a new vector.

Correct



The event list for a sequential procedure can contain level-sensitive events.

- ☒ True
- ☐ False

The event list for a sequential procedure must contain only posedge and negedge events, not level-sensitive events.

Incorrect



What is the effect of mixing blocking and nonblocking assignments in the same procedural block?

- ☒ It has no effect on the simulation.
- ☐ It can lead to race conditions and unpredictable behavior.
- ☐ It improves simulation performance.
- ☐ It simplifies the code.

Mixing blocking and nonblocking assignments in the same procedural block can lead to race conditions and unpredictable behavior.

Incorrect



What is the primary trade-off when implementing clock gating in designs?

- ☒ Reduced performance
- ☐ Increased power consumption
- ☐ Increased clock frequency
- ☐ More gates

Implementing clock gating in RTL design reduces power consumption but requires more gates, affecting the area and performance trade-off.

Incorrect



What are the benefits of defining FSM states using parameters instead of macros?

(Select all that apply)

- ☒ Recognized by synthesis tools for optimization
- ☐ Local scope to the declaring block
- ☐ Global scope across files
- ☐ Easier to redefine from other parts of the design

Parameters are recognized by synthesis tools for FSM optimizations and have a local scope to the declaring block, preventing erroneous use.

Incorrect



What is the result of the expression

```
num = rega & regb;
```

if rega = 4'b1001 and regb = 4'b1010?

- ☐ 4'b1000
- ☐ 4'b1111
- ☐ 4'b0000
- ☒ 4'b1011

The bitwise AND operation between rega and regb results in 4'b1000.

Incorrect

What is the purpose of the \$save and \$restart system tasks in Verilog?

- ☒ To generate clocks
- ☐ To open and close files
- ☐ To save and reload the simulation state
- ☐ To display simulation time

The \$save system task saves the simulation state, and the \$restart system task reloads the simulation state, allowing the simulation to continue from a saved point.

Incorrect

Which operator is used for nonblocking assignments in Verilog?

- ☒ ==
- ☐ =>
- ☐ =
- ☐ <=

The nonblocking assignment operator in Verilog is <=.

Incorrect

What are the key behaviors modeled in low-power simulation?

(Select all that apply)

- ☒ State retention
- ☐ Port isolation
- ☐ State loss
- ☐ Power-up glitches

Low-power simulation models port isolation, state retention, and state loss behaviors to mimic the target hardware's reduced-power behavior.

Incorrect

Which of the following statements about nonblocking assignments are true?
(Select all that apply)

- ☒ They block execution until the assignment completes.
- ☐ They are used primarily in sequential logic.
- ☐ They schedule completion and do not update immediately.
- ☐ They can be mixed with blocking assignments to the same variable in the same procedural block.

Nonblocking assignments schedule completion and do not block further execution, and they are used primarily in sequential logic.

Incorrect



What is the purpose of a reset synchronizer in RTL design?

- ☒ To reduce power consumption
- ☐ To minimize metastability
- ☐ To increase performance
- ☐ To generate clocks

A reset synchronizer minimizes the possibility of metastability when an asynchronous reset signal crosses from one clock domain to another.

Incorrect



X values in clock gating logic can hang a whole chip.

- ☒ True
☐ False

X values in clock gating logic or low-power control can hang a whole chip, making it non-recoverable.

Correct



Which operator is used for logical negation in Verilog?

- ☒ ~
☐ &
☐ !
☐ |

The logical negation operator (!) inverts the logical sense of its operand.

Incorrect



Which event control should be used for modeling combinational logic in an always block?

- ☒ negedge
- ☐ @*
- ☐ posedge
- ☐ @(posedge clk)

The @* wildcard sensitivity list automatically includes all input signals to the logic, ensuring correct simulation results.

Incorrect



Which of the following statements about the if conditional statement in Verilog are true? (Select all that apply)

- ☒ It executes the statement associated with the first known true condition
- ☐ It can only be used with single-bit conditions
- ☐ It tests conditions in sequence
- ☐ Conditions can overlap

The if conditional statement tests conditions in sequence, conditions can overlap, and it executes the statement associated with the first known true condition.

Incorrect



What is the default type of a subroutine port in Verilog?

- ☒ wire
- ☐ real
- ☐ integer
- ☐ reg

A subroutine port is by default a single-bit reg.

Incorrect



What is the primary purpose of using subroutines in Verilog?

- ☒ To encapsulate functionality and make code more readable and reusable
- ☐ To define module parameters
- ☐ To declare variables
- ☐ To perform arithmetic operations

Subroutines encapsulate frequently used code, making it more manageable, readable, and reusable.

Correct



The wait statement blocks if the expression is not true and then unblocks when the expression becomes true.

- ☒ True
☐ False

The wait statement blocks if the expression is not true and then unblocks when the expression becomes true.

Correct



What is the purpose of the proc command in Tcl?

- ☒ To execute commands a specified number of times
☐ To create new Tcl commands
☐ To delete variables
☐ To write to the shell

The proc command is used to create new Tcl commands, which look and behave just like built-in commands.

Incorrect



What does the \$time system function return in Verilog?

- ☒ Simulation time as a 32-bit integer
- ☐ Simulation time as a 64-bit integer
- ☐ Simulation time as a string
- ☐ Simulation time as a real number

The \$time system function returns the simulation time as a 64-bit integer.

Incorrect



Subroutine arguments in Verilog are passed by reference.

- ☒ True
- ☐ False

Subroutine arguments in Verilog are passed by value.

Incorrect



What is the effect of violating setup or hold times in a flip-flop?

- ☒ Stable output
- ☐ Increased performance
- ☐ Predictable delay
- ☐ Metastability

Violating setup or hold times in a flip-flop causes metastability, resulting in unstable and unpredictable output.

Incorrect



Given the following RTL code snippet, what will be the value of `op` if `sel` is unknown (X)?

- ☒ ipa
- ☐ ipb
- ☐ X
- ☐ Z

```
if (sel)
    op = ipa;
else
    op = ipb;
```

In RTL simulation, if `sel` is unknown (X), the condition is treated as other-than-true, and `op` will be assigned the value of `ipb`.

Incorrect



Which of the following are low-power library cells required for implementing the Power Shutoff (PSO) technique? (Select all that apply)

- ☒ Power switches
- ☐ Retention cells
- ☐ AON buffers
- ☐ Isolation cells

PSO design requires power switches, isolation cells, retention cells, and AON buffers for signals that feed through the shutoff domain.

Incorrect



The IEEE Std. 1364.1-2002 defines the synthesizable Verilog constructs.

- ☒ True
- ☐ False

The IEEE Std. 1364.1-2002 defines the synthesizable Verilog constructs.

Correct



What is the primary difference between functions and tasks in Verilog?

- ☒ Functions can have zero or more input/output ports, while tasks must have at least one input port.
- ☐ Tasks return a single value through their name, while functions do not.
- ☐ Tasks can contain timing controls, while functions cannot.
- ☐ Functions can contain timing controls, while tasks cannot.

Tasks can contain timing controls, while functions cannot invoke the scheduler in any way.

Incorrect



The timescale directive specifies the time unit and time precision for subsequent module declarations.

- ☒ True
- ☐ False

The timescale directive specifies the time unit and time precision for subsequent module declarations.

Correct



Which Verilog construct is used to advance simulation time?

- ☒ @ event control
- ☐ always block
- ☐ wait level-sensitive event control
- ☐ # delay

The # delay control is used to advance simulation time in Verilog.

Incorrect



What is the primary trade-off in IC design optimization?

- ☒ Complexity, Efficiency, and Reliability
- ☐ Power, Performance, and Area (PPA)
- ☐ Speed, Size, and Cost
- ☐ Functionality, Usability, and Scalability

IC design optimization involves balancing power, performance, and area to achieve the most economical design.

Incorrect



If $a = 1$, $b = 0$, and $sel = 1$, what will be the value of y after executing the following code snippet?

- ☒ z
- ☐ 1
- ☐ x
- ☐ 0

```
always @(a or b or sel)
begin
    y = a;
    if (sel)
        y = b;
end
```

The nonblocking assignments schedule the updates, ensuring b gets the old value of a , before a is incremented.

Incorrect



Which formal verification technique is used to ensure that clock gating does not change the functionality of the design?

- ☒ Simulation-based verification
- ☐ Timing analysis
- ☐ Power analysis
- ☐ Equivalence checking

Equivalence checking is used to verify that the original design is the same as the design modified to have clock gating, ensuring no change in functionality.

Incorrect



Blocking assignments can lead to race conditions when the same event triggers multiple procedures.

- ☒ True
☐ False

Blocking assignments can lead to race conditions when the same event triggers multiple procedures that execute in a nondeterministic order.

Correct



Which operator performs a bitwise AND operation in Verilog?

- ☒ |
☐ &
☐ ~
☐ &&

The bitwise AND operator (&) performs a logical AND operation on each bit of its operands.

Incorrect



Synthesis tools can optimize the position and number of registers in your design.

- ☒ True
☐ False

Synthesis tools do not optimize the registers or latches in your design; pipeline considerations are your responsibility.

Incorrect



Nonblocking assignments are evaluated using the current values of variables and scheduled for the Non-Blocking Assignment (NBA) region.

- ☒ True
☐ False

Nonblocking assignments are evaluated using the current values of variables and scheduled for the NBA (Nonblocking Assignments) region.

Correct



Which of the following are common uses of scripting in verification?

(Select all that apply)

- ☒ To save and restore debugging environments
- ☐ To control simulation runs for consistency
- ☐ To manually verify the results
- ☐ To compile and build test environments

Scripting is used to compile and build test environments, control simulation runs for consistency, and save and restore debugging environments.

Incorrect

What will be the value of `byte` after executing the following code snippet?

- ☒ 8'b11111111
- ☐ 8'b00001111
- ☐ 8'b00000000
- ☐ 8'b11110000

```
reg [7:0] byte = 8'b00001111;  
initial begin  
    byte[3:0] = byte[7:4];  
    byte[7:4] = byte[3:0];  
end
```

The blocking assignments cause the upper half to be assigned to the lower half first, resulting in both halves having the same value, which leads to `byte` being 8'b00000000.

Incorrect

What is the primary purpose of using the `expr` command in Tcl?

- ☒ To create new Tcl commands
- ☐ To execute commands a specified number of times
- ☐ To delete variables
- ☐ To evaluate mathematical expressions

The `expr` command is used to evaluate mathematical expressions in Tcl.

Incorrect



What is the purpose of using one-hot encoding in FSM optimization?

- ☒ To simplify output decode logic
- ☐ To simplify timing constraints
- ☐ To increase noise
- ☐ To increase power consumption

One-hot state encoding reduces the combinational logic required to encode and decode the state.

Correct



Which command in Tcl is used to write to the shell?

- ☒ unset
- ☐ proc
- ☐ set
- ☐ puts

The puts command is used to write to the shell in Tcl.

Incorrect

What will be the value of b after executing the following code snippet?

- ☒ The value of b will be unpredictable.
- ☐ The value of b will be the incremented value of a.
- ☐ The value of b will be the old value of a.
- ☐ The value of b will be zero.
- ☐ The value of b will be 1'bx.

```
initial begin
    a = 0;
    b = 0;
end
always @(posedge
always @(posedge
```

The blocking assignments can lead to race conditions, making the value of b unpredictable.

Correct

Which physical issue can cause functional errors due to glitches in the crossover path?

- ☒ X-Propagation
- ☐ Power consumption
- ☐ Clock Domain Crossing
- ☐ Area reduction

Clock Domain Crossing can cause functional errors due to glitches in the crossover path.

Incorrect



The \$write system task in Verilog appends a newline character at the end of the output.

- ☒ True
- ☐ False

The \$write system task does not append a newline character at the end of the output, unlike the \$display system task.

Incorrect



What is the advantage of using tasks for stimulus generation in Verilog?

- ☒ Tasks can contain delays and timing events
- ☐ Tasks can initialize the simulation
- ☐ Tasks execute in zero time
- ☐ Tasks return a single value

Tasks enhance readability and encapsulate repeated sequences of control signals, making the code more compact and easier to manage.

Correct

Which of the following statements about modeling sequential logic are true?

(Select all that apply)

- ☒ Model set/reset behaviors in the first branches of an if statement
- ☐ The event list must contain only posedge and negedge events
- ☐ Include level-sensitive events in the event list
- ☐ Use blocking assignments for storage variables

For modeling sequential logic, the event list must contain only posedge and negedge events, and set/reset behaviors should be modeled in the first branches of an if statement. Use nonblocking assignments for storage variables and avoid level-sensitive events in the event list.

Incorrect

Which command is used to create a variable and assign/retrieve its value in Tcl?

- ☒ proc
- ☐ set
- ☐ unset
- ☐ puts

The set command is used to create a variable and assign/retrieve its value in Tcl.

Incorrect



What is the purpose of the Unified Power Format (UPF) language in low-power design?

- ☒ To specify power intent
- ☐ To describe the functionality of the design
- ☐ To optimize performance
- ☐ To reduce area

The Unified Power Format (UPF) language is used to specify power intent in low-power design, addressing techniques related to physical effects.

Correct



Which directive is used to include the contents of a file in Verilog?

- ☒ ``include`
- ☐ ``ifdef`
- ☐ ``define`
- ☐ ``timescale`

The ``include` directive is used to insert the contents of a file at the point where the directive appears.

Correct

In which type of procedural block should you primarily use nonblocking assignments?

- ☒ Combinational logic
- ☐ Initial blocks
- ☐ Sequential logic
- ☐ Continuous assignments

Nonblocking assignments should be used primarily in procedural blocks that represent sequential logic.

Incorrect

What is the purpose of the \$timeformat system task in Verilog?

- ☒ To get the simulation time
- ☐ To monitor changes in signal values
- ☐ To write values to a file
- ☐ To set the display format for the %t formatter

The \$timeformat system task sets the display format for the %t formatter, controlling units, precision, suffix, and field width.

Incorrect



What is the purpose of the info command in Tcl?

- ☒ To write to the shell
- ☐ To delete one or more variables
- ☐ To create a variable
- ☐ To retrieve information about Tcl commands and variables

The info command is used to retrieve information about Tcl commands and variables.

Incorrect



What is the primary function of the \$display system task in Verilog?

- ☒ To write values to a file
- ☐ To print values to the standard output with a newline
- ☐ To monitor changes in signal values
- ☐ To format time display

The \$display system task prints values to the standard output and appends a newline character at the end.

Incorrect



Incomplete assignments in a combinational procedure infer a latch.

- ☒ True
- ☐ False

If an execution path through a combinational procedure exists that does not update the value of some output, the synthesis tool infers a latch to implement this behavior.

Correct



The synthesis tool infers three-state logic when you assign a net or variable the high-impedance value (z).

- ☒ True
☐ False

Synthesis infers three-state logic when you assign a net or variable the high-impedance value (z).

Correct



Which system task would you use to continuously monitor nets and variables in Verilog?

- ☒ \$display
☐ \$write
☐ \$monitor
☐ \$strobe

The \$monitor system task continuously monitors its arguments and displays their values at the end of the time slice in which any argument changes.

Incorrect



The `timescale directive must appear inside a module.

- ☒ True
- ☐ False

The `timescale directive must appear outside of the module.

Incorrect



What is the primary difference between a Moore machine and a Mealy machine?

- ☒ Mealy machine outputs depend on the previous state.
- ☐ Moore machine outputs depend on the current state and inputs.
- ☐ Mealy machine outputs depend only on the current state.
- ☐ Moore machine outputs depend only on the current state.

In a Moore machine, the outputs are a function solely of the current state vector, whereas in a Mealy machine, the outputs depend on the current state and at least one current input.

Incorrect



Using the wildcard event control (@*) in Verilog ensures that all inputs to a combinational procedure are included in the event list.

- ☒ True
☐ False

The wildcard event control (@*) ensures that all inputs to the procedure are included in the event list, avoiding incorrect RTL simulation results.

Correct



What is the primary purpose of using the source command in Tcl?

- ☒ To create new Tcl commands
☐ To write to the shell
☐ To delete variables
☐ To execute commands from a file

The source command is used to execute commands from a file in Tcl.

Incorrect

